

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
to find the current drawn by the head light bulb , we use the formula $I = P/V$,
where I is current , P is power , and V is voltage .
so $I = 40 / 12 = 3.33$ amps
using the formula $R = V/I$, we can find the resistance .
 $R = 12 / 3.33 = 3.6$ oh ms
so the head light bulb will draw a current of 3.33 amps and its resistance must be 3.6 oh ms .
</think><answer> 3.33 amps , 3.6 oh ms

